Theory of Everything (operator & readout): Unitary core, ABS-first measurement, moral semigroup, legacy invariant, and auditable witnesses

Preface (what this document *is*)

This is a *fully self-contained operator/readout theory* with explicit definitions and internal theorems. It is written "top-down": (I) the unified theory, (II) specializations/modules, (III) code appendix.

Important boundary. This document does *not* claim unconditional proofs of the Millennium Problems (RH, P vs NP, Navier–Stokes). Instead, it provides:

- a rigorous pipeline in which those problems naturally reappear as *stress tests* of projection, witness, and stability,
- explicit lemmas you can *prove* (tail bounds, projection stability, semigroup irreversibility, witness interfaces),
- a place to attach any future proof as an additional module without rewriting the core.

Contents

1 The Unified Theory (core, no fragments)

1.1 Primitive objects

Definition 1.1 (Latent state space)**.** A *system* is a complex Hilbert space $\mathcal{H}$ with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$. A *latent state* is $\psi(t) \in \mathcal{H}$ for $t \in \mathbb{R}$.

Definition 1.2 (Readout space and observable)**.** A *readout* is a measurable scalar map $A : \mathcal{H} \to \mathbb{R}_{\geq 0}$. A canonical class is **ABS-first** readout:

$$A(\psi) = |M(\psi)| \quad \text{for a bounded linear functional } M \in \mathcal{H}^*, \qquad \text{or} \qquad A(\psi) = \|M\psi\| \text{ for } M \in \mathcal{B}(\mathcal{H}). \tag{1}$$

Definition 1.3 (Four-sector decomposition)**.** Fix orthogonal projections $P_1, \ldots, P_4$ on $\mathcal{H}$ such that

$$P_i^2 = P_i, \quad P_i^* = P_i, \quad P_iP_j = 0 \ (i \neq j), \quad \sum_{i=1}^{4} P_i = I. \tag{2}$$

The *four sectors* are $\mathcal{H}_i := P_i\mathcal{H}$.

1.2 Core dynamics: conservative interior + nonlinear dashboard

Axiom 1.4 (Unitary core)**.** There exists a strongly continuous unitary family $U(t)$ on $\mathcal{H}$ such that

$$\psi(t) = U(t)\psi(0), \qquad U(t)^*U(t) = I \quad (\forall t). \tag{3}$$

Proposition 1.5 (No pump/no compressor inside)**.** *Under Axiom 3, $\|\psi(t)\| = \|\psi(0)\|$ for all t.*

Proof. Unitarity implies $\|U(t)\psi(0)\| = \|\psi(0)\|$. □

Remark 1.6. Large apparent "spikes" are allowed in *readout* $A(\psi(t))$ because $\psi \mapsto A(\psi)$ can be nonlinear (ABS, coarse-graining, bounded observables, etc.), even when the interior is norm-preserving.

1.3 Where "chaos" can live (three standard placements)

Definition 1.7 (Three placements)**.** Let $U(t)$ be unitary. Irregular/chaotic-looking time series can appear via:

1. **Chaotic readout** (projection/coarse-graining): $A(t) = |\langle\phi, M\psi(t)\rangle|$ or $A(t) = \|M\psi(t)\|$.
2. **Chaotic drive in a self-adjoint generator**: $H(t) = H_0 + \epsilon(t)V$ with $H(t) = H(t)^*$ and $\epsilon(t)$ produced by a deterministic chaotic signal.
3. **Open-system noise** (non-unitary on reduced states): Lindblad evolution for $\rho(t)$.

1.4 Gate: admissibility as a *domain cut*

Definition 1.8 (Admissible futures and absorbing state)**.** Let Ω be a set of *possible futures/trajectories*. Fix a forbidden subset $B \subset \Omega$ ("red-line violation") and introduce an absorbing cemetery state $\perp$. Define the moral projection

$$\Pi : \Omega \to \Omega \cup \{\perp\}, \qquad \Pi(\omega) = \begin{cases} \omega, & \omega \notin B, \\ \perp, & \omega \in B. \end{cases} \tag{4}$$

Definition 1.9 (Moral semigroup (irreversibility))**.** Let $T_t : \Omega \to \Omega$ be an underlying evolution map. Define the induced *moral evolution*

$$T_t^{\text{moral}} := \Pi \circ T_t. \tag{5}$$

If $\perp$ is absorbing ($T_t(\perp) = \perp$), then $\{T_t^{\text{moral}}\}_{t\geq 0}$ forms a semigroup:

$$T_{t+s}^{\text{moral}} = T_t^{\text{moral}} \circ T_s^{\text{moral}}, \qquad t, s \geq 0. \tag{6}$$

Remark 1.10. This is the precise meaning of "for some actions there is no future": it is not an energy term, it is a *domain admissibility cut.*

1.5 Zeno-limit enforcement: continuous monitoring as a limit

Definition 1.11 (Zeno-limited evolution (operator form))**.** Let Π_{good} be an orthogonal projection in $\mathcal{H}$ representing the admissible subspace. Define

$$U_Z(t) := \lim_{n\to\infty} \Big(\Pi_{\text{good}}\, U(t/n)\, \Pi_{\text{good}}\Big)^n, \tag{7}$$

whenever the strong limit exists.

Remark 1.12. Equation (7) is a standard "enforce-at-every-step" formalization: morality is implemented as a *constraint on evolution / domain*, not a late audit.

1.6 Legacy as a monotone invariant

Definition 1.13 (Integrity density and legacy)**.** Let $Q(t) \geq 0$ be an integrity density along a trajectory. Define the legacy functional

$$\mathcal{L}(t) := \int_0^t Q(\tau)\, d\tau. \tag{8}$$

Impose monotonicity: if $t_2 \geq t_1$ then $\mathcal{L}(t_2) \geq \mathcal{L}(t_1)$.

Definition 1.14 (Absorbing rule for legacy)**.** If $\omega \in B$ (forbidden), enforce $\mathcal{L}(\omega) = 0$ (legacy is annihilated by domain violation).

1.7 Auditable truth: SAT/UNSAT witness interface

Definition 1.15 (Moralized constraint system)**.** Let Φ_{phys} be the physical/consistency constraints and Φ_{moral} the admissibility constraints. Define

$$\Phi := \Phi_{\text{phys}} \wedge \Phi_{\text{moral}}. \tag{9}$$

A *model* is an assignment σ satisfying $\sigma \models \Phi$. If no such σ exists, then Φ is *UNSAT* and admits a machine-verifiable certificate (proof object).

Remark 1.16. This turns "good vs evil" into an explicit clause-set. The point is not theology; the point is auditability.

2 Modules (parts derived from the core)

2.1 Prime-log spectrum and ABS-safe interference (RH-facing module)

Proposition 2.1 (Prime frequencies)**.** *For* $\operatorname{Re}(s) > 1$*,*

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \qquad \log\zeta(s) = \sum_p \sum_{r\geq 1} \frac{1}{r} p^{-rs}. \tag{10}$$

With $s = \sigma + it$, the oscillatory factors have angular frequencies $\omega_{p,r} = r\log p$.

Definition 2.2 (Sagnac-like ABS readout)**.** Fix weights a_p, b_p and a cutoff P:

$$S_+(t) = \sum_{p\leq P} a_p e^{it\log p}, \qquad S_-(t) = \sum_{p\leq P} b_p e^{-it\log p}. \tag{11}$$

Define ABS-safe interference observables

$$A_\pm(t) = |S_+(t) \pm S_-(t)|\,. \tag{12}$$

Remark 2.3. Zeros are amplitude nodes; phase becomes ill-conditioned at nodes. ABS observables avoid branch singularities. This module is designed to be numerically stable and falsifiable as a *readout* construction.

2.2 Explicit tail/leakage control (bank-kernel module)

Definition 2.4 (Bank Gram kernel)**.** Assume a kernel decomposes as

$$G_{\text{bank}}(\Delta) = \sum_{j=0}^{J} w_j^2 K_j(\Delta). \tag{13}$$

Assume polynomial decay: for some integer $N \geq 4$ there exist $C_{j,N} > 0$ such that

$$|K_j(\Delta)| \leq C_{j,N}(1 + |\Delta|)^{-N} \qquad (\Delta \in \mathbb{R}). \tag{14}$$

Lemma 2.5 (Tail constant bound)**.** *Define*

$$T(J) := \sum_{m \geq 1} (m+1) \sup_{|\Delta| \in [m,m+1]} |G_{\text{bank}}(\Delta)| . \tag{15}$$

Then

$$T(J) \leq (\zeta(N-1) - 1) \sum_{j=0}^{J} w_j^2 C_{j,N}. \tag{16}$$

Proof. Use the majorant $|G_{\text{bank}}(\Delta)| \leq (\sum_j w_j^2 C_{j,N})(1 + |\Delta|)^{-N}$ and $\sup_{|\Delta| \in [m,m+1]}(1 + |\Delta|)^{-N} \leq (m+1)^{-N}$, then reindex the zeta tail: $\sum_{m \geq 1}(m+1)^{-(N-1)} = \sum_{k \geq 2} k^{-(N-1)} = \zeta(N-1) - 1$. $\square$

Proposition 2.6 (Checkable leakage inequality)**.** *If a leakage functional obeys*

$$\theta(t; J) \leq \frac{(\log t)\, T(J)}{G_{\text{bank}}(0)} \quad \textit{with } G_{\text{bank}}(0) > 0, \tag{17}$$

then $\theta(t; J) < 1$ is guaranteed by the explicit inequality

$$\sum_{j=0}^{J} w_j^2 C_{j,N} < \frac{G_{\text{bank}}(0)}{(\log t)(\zeta(N-1) - 1)}. \tag{18}$$

2.3 4D information block (the operational definition)

Definition 2.7 (4D information block)**.** A *4D information block* is a tuple

$$\mathcal{B}_{4D} := (\psi(t), U(t), G, \mathcal{R}), \tag{19}$$

where $\psi(t) \in \mathcal{H}$ (latent state), $U(t)$ is the interior evolution, G is the Gate/admissibility operator (domain cut and/or Zeno enforcement), and $\mathcal{R}$ is the ABS-first readout producing auditable scalars and witnesses.

Remark 2.8. This definition is intentionally minimal: it is what you actually *run*, measure, and publish.

3 Research interfaces (RH, P vs NP, Navier–Stokes)

Boundary statement

The following are *interfaces*, not claims of solved problems. They explain how the unified pipeline connects to each domain.

3.1 RH interface

RH is cast as a statement about zeros of $\zeta(s)$. In this framework:

- the interior object is analytic (spell),
- the numerics is a projection/readout (dashboard),
- stability is enforced by tail/leakage bounds and ABS-safe observables (gate).

A genuine RH proof would require a global analytic/spectral mechanism beyond readout stability.

3.2 P vs NP interface

SAT instances are constraint systems Φ. In this framework:

- *search* lives in a huge latent space (internal superposition),
- *verification* is a projected witness (DIMACS / certificate),
- morality is an extra clause family Φ_{moral}.

A genuine separation proof is a complexity-theoretic result not supplied here.

3.3 Navier–Stokes interface

NS regularity is the question of whether solutions remain admissible (smooth, finite) for all time. In this framework:

- the interior dynamics is nonlinear,
- coarse observation makes deterministic flow look chaotic,
- the "Gate" corresponds to an a priori bound preventing blow-up.

A genuine proof would require a global control estimate; this document supplies a template language for such gates.

4 Code appendix (implementation contract)

4.1 Deterministic seed: `n_singular`

Definition 4.1 (Seed parameter)**.** Let $n_{\text{singular}} \in \mathbb{N}$ be a deterministic seed that drives reproducible construction of:

- hash-based bitfields (quantized projection),
- SAT clauses / variable ordering (auditable witness path),
- parameter schedules (e.g. $\Delta(t)$, window banks, w_j).

4.2 Minimal implementation contract

Your Python must expose (at least) the following functions:

- `seed_from_n_singular(n_singular) -> bytes/int`
- `build_bank(seed) -> (w, K, C_jN, G0)`
- `compute_theta(t, w, C_jN, G0, N) -> float` implementing (18)
- `moral_gate(action/state) -> bool or ABSORB`
- `emit_dimacs(physics, moral) -> cnf`
- `solve_sat(cnf) -> (SAT/UNSAT, witness)`

4.3 Where to wire the new rule (single point of truth)

- **One** canonical place defines n_{singular} and derives the seed.
- **One** canonical place defines the moral gate and produces Φ_{moral} (or absorbing $\perp$).
- All scoring/robustness modules must read these (never re-derive with inconsistent conventions).

End note

If you want, the next step is to split this master into:

1. `toe_core.tex` (sections 1–2),
2. `toe_modules.tex` (section 3),
3. `toe_code.tex` (section 4),

and compile via `\input` so you can version-control each layer independently.